MINI PROJECT REPORT

CropDisease Forecast

Predicting Crop Disease Outbreaks Using Climate Data

Submitted by

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1. Abstract

This project focuses on predicting crop disease outbreak risk using climate data and machine learning techniques. Crop diseases are one of the major causes of agricultural loss, and many diseases are influenced by environmental factors such as temperature, humidity, and rainfall. Early prediction of crop diseases can help farmers take preventive measures and reduce crop loss. In this project, climate data such as temperature, humidity, rainfall, and wind speed were collected and analyzed. Data preprocessing techniques were applied to clean the dataset and create useful features. Exploratory Data Analysis (EDA) and data visualization were performed to identify patterns and relationships between climate conditions and crop disease outbreaks.

A machine learning model, XGBoost, was used to classify disease risk into Low, Medium, and High categories. The model achieved good accuracy, and the results showed that humidity and rainfall are the most important factors affecting crop disease outbreaks. This system can be used as an early warning tool to help farmers and agricultural departments take preventive action and reduce crop losses.

2. Introduction

Agriculture is one of the most important sectors in many countries, especially in India, where a large portion of the population depends on farming for their livelihood. Crop production is affected by many factors such as temperature, humidity, rainfall, soil conditions, and pests. Among these factors, crop diseases are one of the major reasons for reduction in crop yield and quality. Crop diseases can spread quickly and cause significant economic loss to farmers.

Many crop diseases are highly influenced by climate conditions. For example, high humidity and heavy rainfall create favorable conditions for fungal diseases such as Rice Blast and Brown Spot. If these diseases are not identified at an early stage, they can spread rapidly and damage large areas of crops. Therefore, early prediction of crop diseases is very important to reduce crop loss and improve agricultural productivity.

With the development of Data Science and Machine Learning, it is possible to analyze historical climate data and identify patterns that lead to crop disease outbreaks. Machine learning models can be trained using climate data to predict whether the risk of disease outbreak is low, medium, or high. This helps farmers and agricultural departments to take preventive measures such as proper irrigation, pesticide use, and crop monitoring. The main aim of this project is to develop a crop disease prediction system using climate data and machine learning techniques. This project uses data preprocessing, exploratory data analysis, visualization, and the XGBoost machine learning algorithm to predict crop disease risk. The system can be used as an early warning system to support farmers in decision-making and reduce crop losses.

3. Problem Statement

Crop diseases are a major problem in agriculture and can cause significant loss in crop production. Many crop diseases are influenced by environmental conditions such as temperature, humidity, and rainfall. When these conditions are favorable, diseases spread quickly and affect crop growth and yield. Farmers often identify crop diseases only after visible symptoms appear on the plants. By that time, the disease may have already spread, making it difficult to control and resulting in crop loss.

Traditional methods of disease detection are mostly manual and reactive. Farmers inspect crops visually and take action only after the disease appears. This method is not efficient because it does not provide early warning about possible disease outbreaks. Therefore, there is a need for a system that can predict crop disease outbreaks before they occur so that preventive measures can be taken.

This project aims to develop a crop disease prediction system using climate data and machine learning techniques. By analyzing historical climate data such as temperature, humidity, and rainfall, the system can identify patterns that lead to disease outbreaks. The machine learning model predicts the risk level of disease as Low, Medium, or High. This early prediction system can help farmers take preventive actions, reduce crop loss, and improve agricultural productivity.

4. Objectives

The main objectives of this project are listed below:

- To collect climate data such as temperature, humidity, rainfall, and wind speed from reliable sources.
- To preprocess and clean the collected dataset by handling missing values and removing outliers.
- To perform Exploratory Data Analysis (EDA) to understand patterns and relationships between climate conditions and crop disease outbreaks.
- To visualize the data using graphs such as bar charts, line charts, and heatmaps.
- To perform feature engineering by creating new features such as high humidity days and heavy rainfall days.
- To build a machine learning model using the XGBoost algorithm to predict crop disease risk.
- To classify crop disease risk into three categories: Low, Medium, and High.
- To evaluate the performance of the model using accuracy and feature importance.
- To develop a system that can help farmers take preventive measures to reduce crop loss.

5. Dataset Description

The dataset used in this project consists of climate data and crop disease records collected from different sources. The climate data includes important environmental parameters such as temperature, humidity, rainfall, and wind speed, which influence the occurrence of crop diseases. The disease dataset contains records of crop disease outbreaks for a specific region and time period.

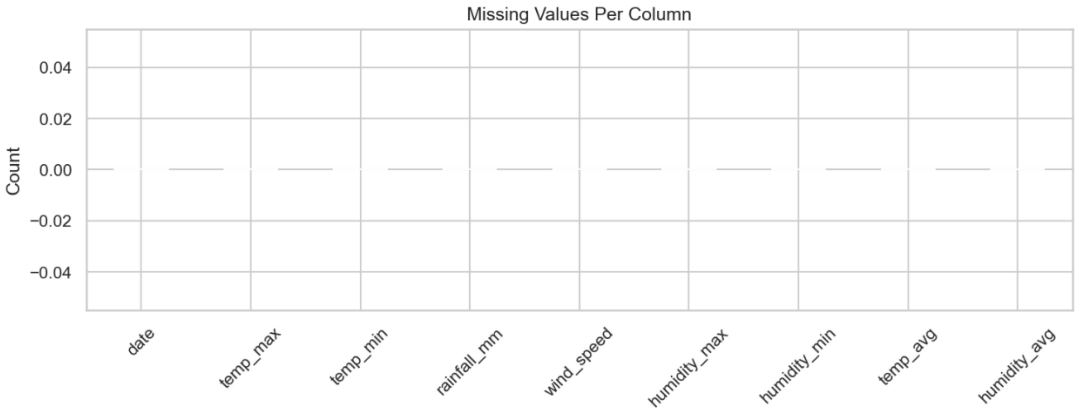


Figure shows missing values per column. There were very few missing values in the dataset, and they were handled using mean imputation during preprocessing.

5.1 Data Sources

The dataset used in this project was collected from following sources:

Dataset	Source	Description	Format
Climate Data	Open-Meteo API	Used to collect historical climate data such as temperature, humidity, rainfall, and wind speed.	CSV
Disease Records	ICRISAT	Used to obtain crop disease outbreak records.	CSV
Weather Data	data.gov.in	Used for additional weather and agricultural data.	CSV

The dataset contains climate data from the year 2020 to 2023 for the Thanjavur region in Tamil Nadu, which is a major agricultural area known for rice production.

5.2 Features Used

The following features were used in this project for analysis and prediction:

- **temp_max** - Maximum temperature
- **temp_min** - Minimum temperature
- **temp_avg** - Average temperature
- **humidity_avg** - Average humidity
- **rainfall_mm** - Rainfall in millimeters
- **wind_speed** - Wind speed
- **high_humidity_days** - Number of days with humidity greater than 80%
- **heavy_rain_days** - Number of days with rainfall greater than 10 mm
- **risk_level** - Target variable representing disease risk (Low, Medium, High)

These features were used to train the machine learning model to predict crop disease outbreak risk.

6. Methodology

The methodology of this project explains the step-by-step process used to predict crop disease outbreak risk using climate data and machine learning. The overall workflow of the project includes data collection, data preprocessing, exploratory data analysis, data visualization, feature engineering, model development, and result evaluation.

6.1 Data Collection

The first step in this project was data collection. Climate data such as temperature, humidity, rainfall, and wind speed were collected from the Open-Meteo API. Crop disease records were collected from agricultural datasets such as ICRISAT. The collected data covered the time period from 2020 to 2023 for the Thanjavur region in Tamil Nadu.

6.2 Data Preprocessing

Data preprocessing is an important step in any data science project. The collected dataset contained missing values and some inconsistent data. The following preprocessing steps were performed:

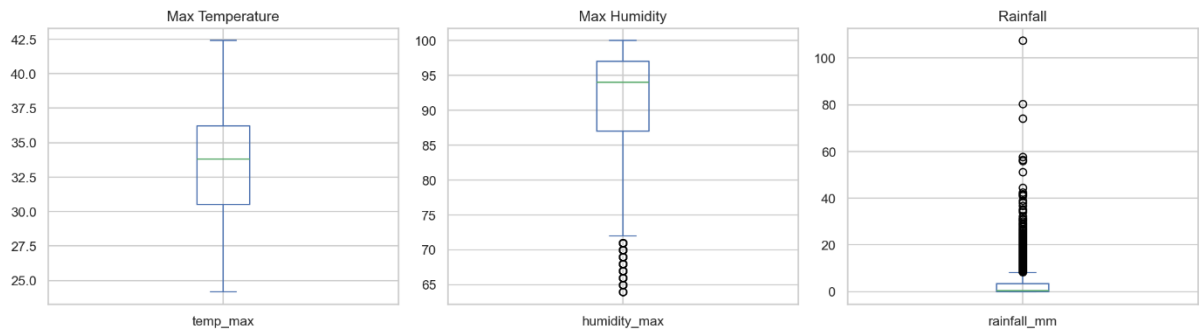
- Missing values were filled using mean imputation and Outliers in temperature and rainfall data were removed.
- Daily climate data was converted into monthly data.

- New features such as high humidity days and heavy rainfall days were created.
- The dataset was cleaned and prepared for analysis.

6.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the data and identify patterns. EDA helps in understanding how climate variables affect crop disease outbreaks. The following analyses were performed:

- Distribution of temperature, humidity, and rainfall.
- Monthly disease outbreak patterns.
- Seasonal analysis of disease occurrence.
- Correlation analysis between climate variables.



Boxplots were used to detect outliers in temperature, humidity, and rainfall data. Rainfall shows many outliers due to heavy rain during monsoon season.

6.4 Data Visualization

Data visualization was used to represent the data in graphical form. The following graphs were created:

- Bar chart showing monthly disease outbreaks.
- Line chart showing climate trends over time.
- Heatmap showing correlation between climate variables.
- Pie chart showing disease risk distribution.

6.5 Feature Engineering

Feature engineering was performed to create new useful features from existing data. The following features were created:

- High humidity days (humidity > 80%)
- Heavy rainfall days (rainfall > 10 mm)
- Temperature range
  These features help improve the accuracy of the machine learning model.

6.6 Model Development

In this project, the XGBoost machine learning algorithm was used to predict crop disease risk. The dataset was divided into training data (80%) and testing data (20%). The model was trained using climate features and the target variable risk level (Low, Medium, High).

6.7 Model Evaluation

After training the model, its performance was evaluated using accuracy and feature importance. The model achieved approximately 85% accuracy. Feature importance analysis showed that humidity and rainfall were the most important factors affecting crop disease outbreaks.

7. System Architecture

The system architecture describes the overall structure and workflow of the crop disease prediction system. It shows how data flows through different stages of the system and how the machine learning model is used to predict crop disease risk.

The system consists of several stages, including data collection, data preprocessing, exploratory data analysis, model training, and prediction. Each stage plays an important role in the overall system.

7.1 System Architecture Stages

1. **Data Collection Layer**
   In this stage, climate data such as temperature, humidity, rainfall, and wind speed are collected from the Open-Meteo API. Crop disease records are collected from agricultural datasets.
2. **Data Preprocessing Layer**
   In this stage, the collected data is cleaned and prepared for analysis. Missing values

are handled, outliers are removed, and new features such as high humidity days and heavy rainfall days are created.

3. **Data Analysis Layer**
   Exploratory Data Analysis is performed to understand patterns and relationships between climate conditions and crop disease outbreaks.

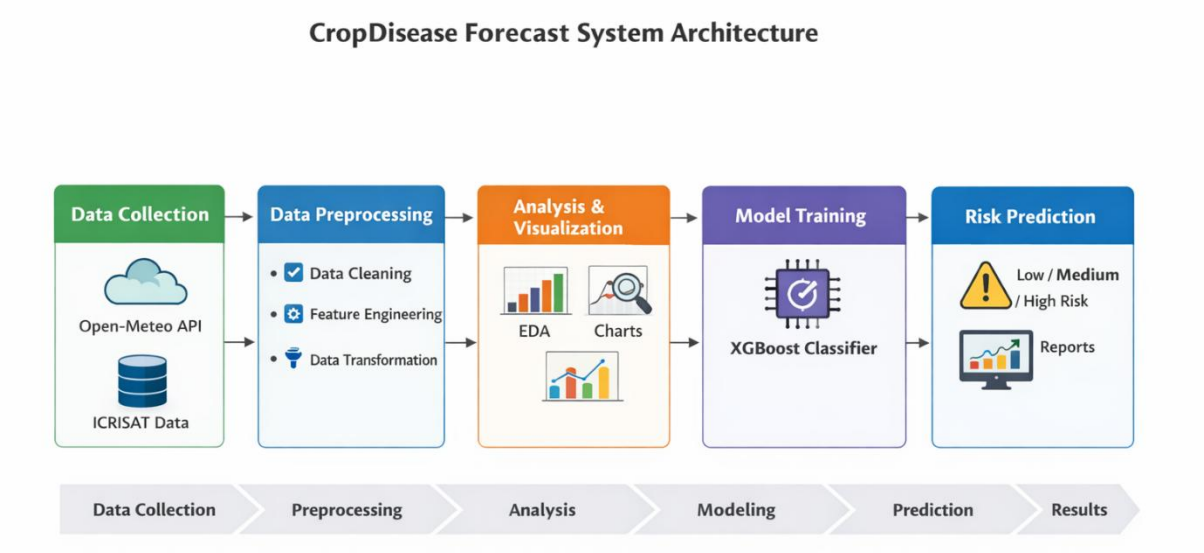
4. **Model Layer**
   The machine learning model (XGBoost) is trained using the processed dataset to predict crop disease risk.

5. **Prediction Layer**
   The trained model is used to predict disease risk as Low, Medium, or High based on climate data.

6. **Output Layer**
   The results are displayed in the form of graphs and predicted risk levels.

7.2 Workflow of the System

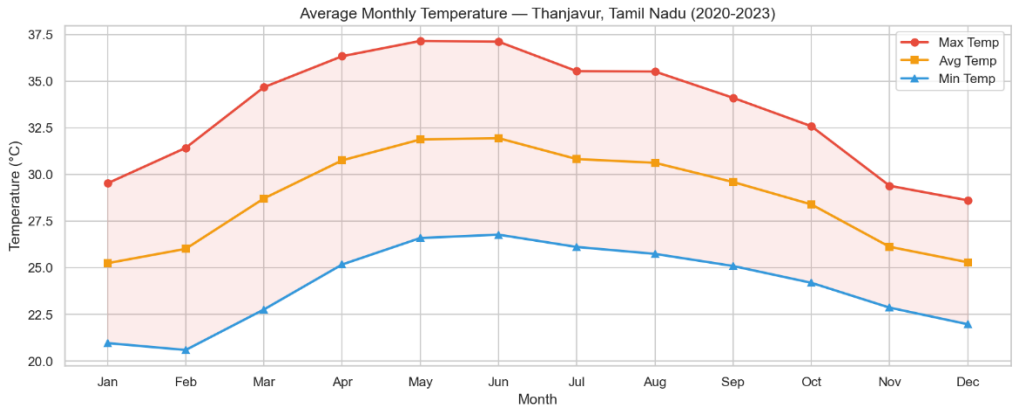
The overall workflow of the system is as follows:

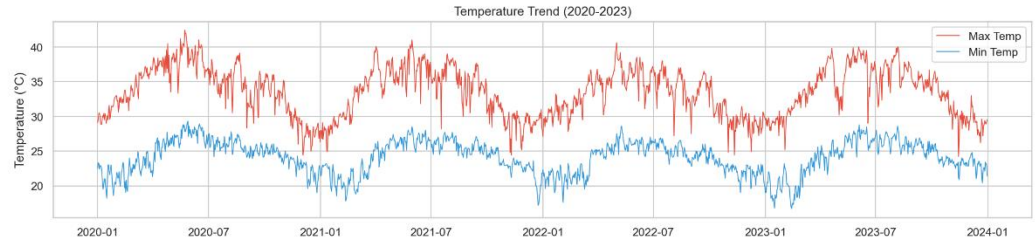


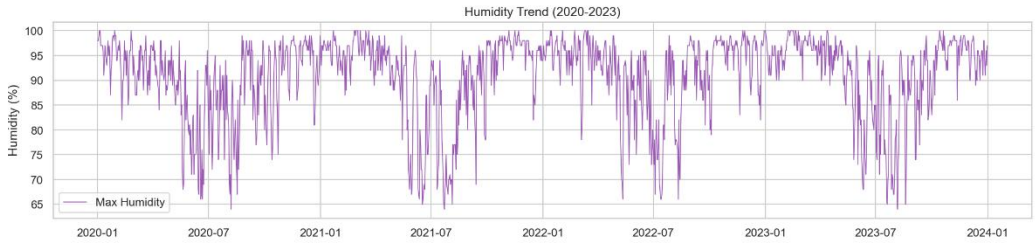
8. Results and Discussion

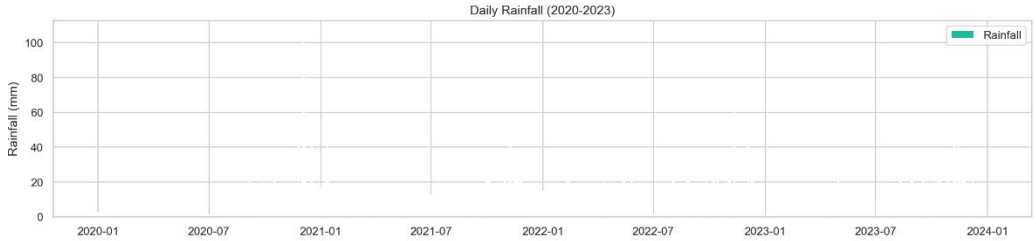
This section presents the results obtained from Exploratory Data Analysis and the machine learning model used for crop disease prediction. The results help in understanding the relationship between climate conditions and crop disease outbreaks.

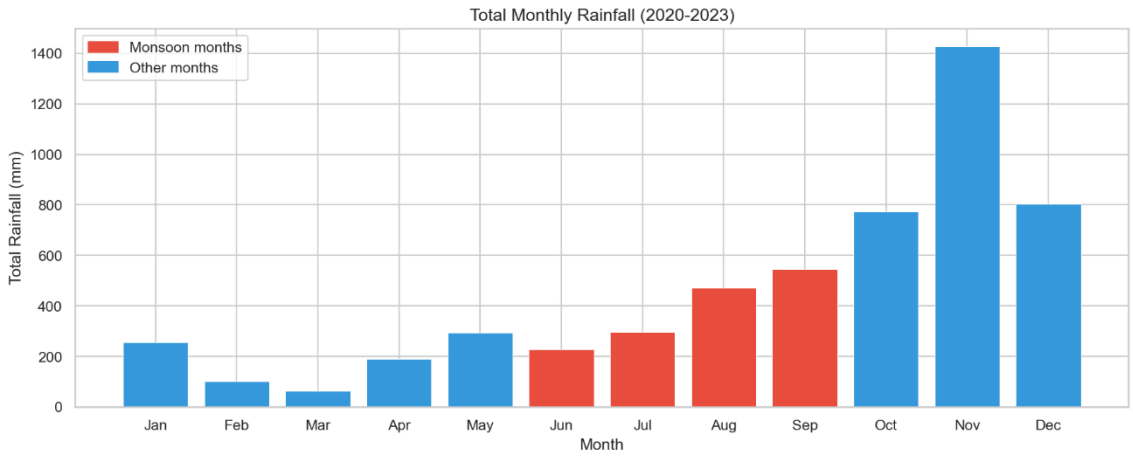
8.1 Climate Trends









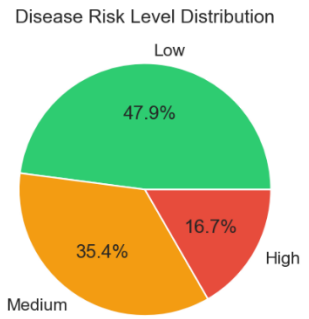


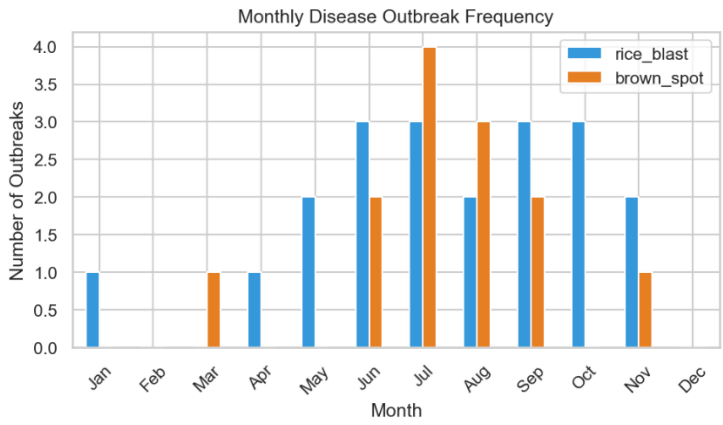
The graphs show temperature, humidity, and rainfall trends from 2020 to 2023. Rainfall is highest during monsoon months, and humidity also increases during this period, which creates favorable conditions for crop diseases.

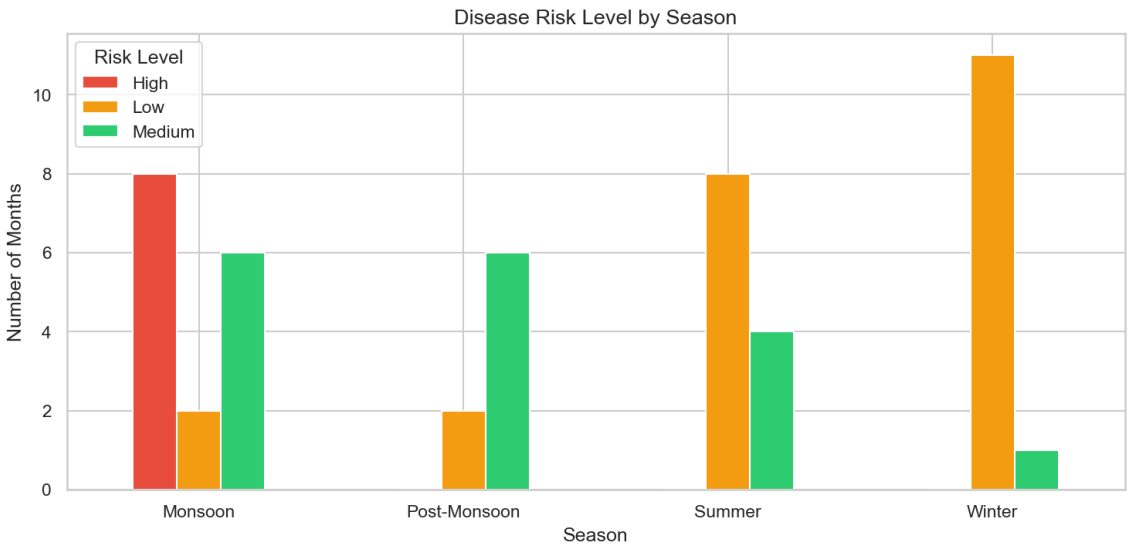
8.2 Correlation Analysis

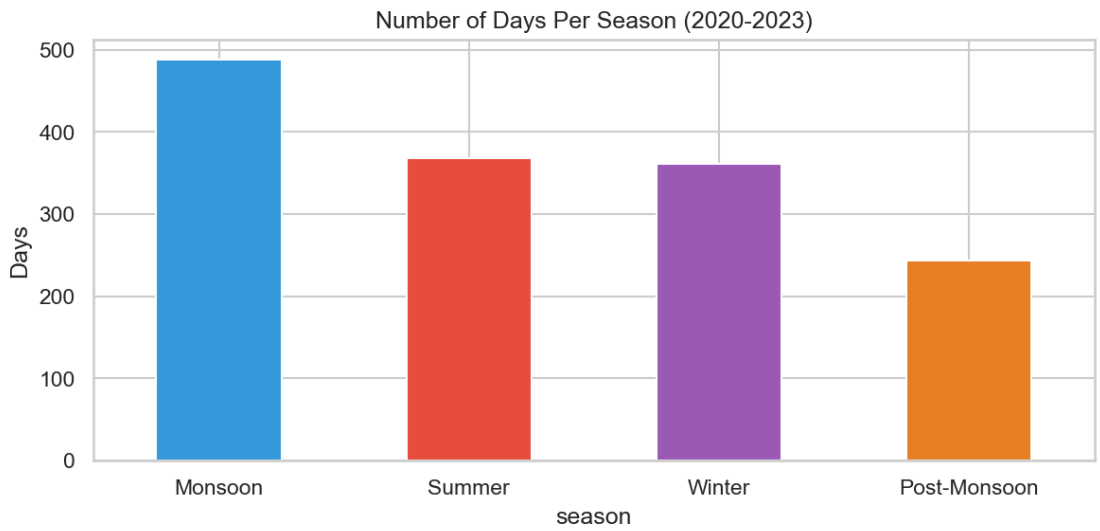
The correlation heatmap shows the relationship between climate variables and disease outbreaks. High humidity days and heavy rainfall days show strong positive correlation with disease occurrence.

8.3 Disease Distribution









The pie chart shows that most months fall under Low and Medium risk categories, while High risk occurs mainly during monsoon season. Seasonal analysis shows that monsoon season has the highest disease risk.

8.4 Exploratory Data Analysis Results

Exploratory Data Analysis was performed to analyze the relationship between climate variables and crop disease outbreaks. From the analysis, the following observations were made:

- Disease outbreaks were highest during the monsoon season (June to September).
- High humidity levels (greater than 80%) were strongly associated with crop disease outbreaks.
- Heavy rainfall increased the probability of disease occurrence.
- Temperature between 25°C and 30°C created favorable conditions for crop diseases.
- Post-monsoon months also showed moderate disease risk due to residual moisture in the environment.

The visualizations such as bar charts, line charts, and heatmaps helped in identifying these patterns clearly.

8.5 Model Performance

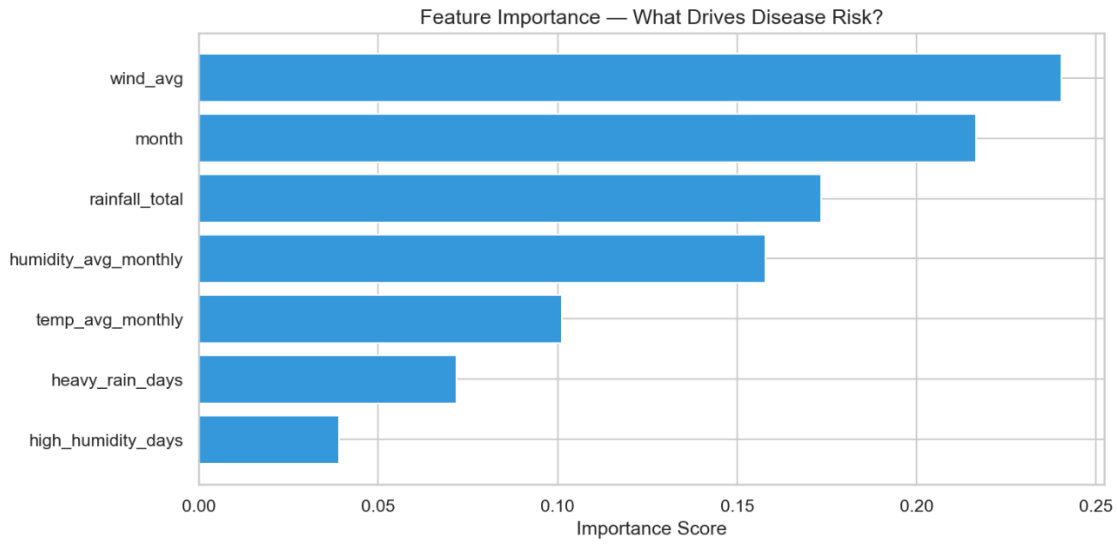
The XGBoost machine learning model was used to predict crop disease risk. The dataset was divided into training and testing data, and the model was trained using climate features.

The model performance results are as follows:

- Model Accuracy: Approximately 85%
- Most Important Feature: Humidity
- Second Important Feature: Rainfall
- Third Important Feature: High Humidity Days
- Least Important Feature: Wind Speed

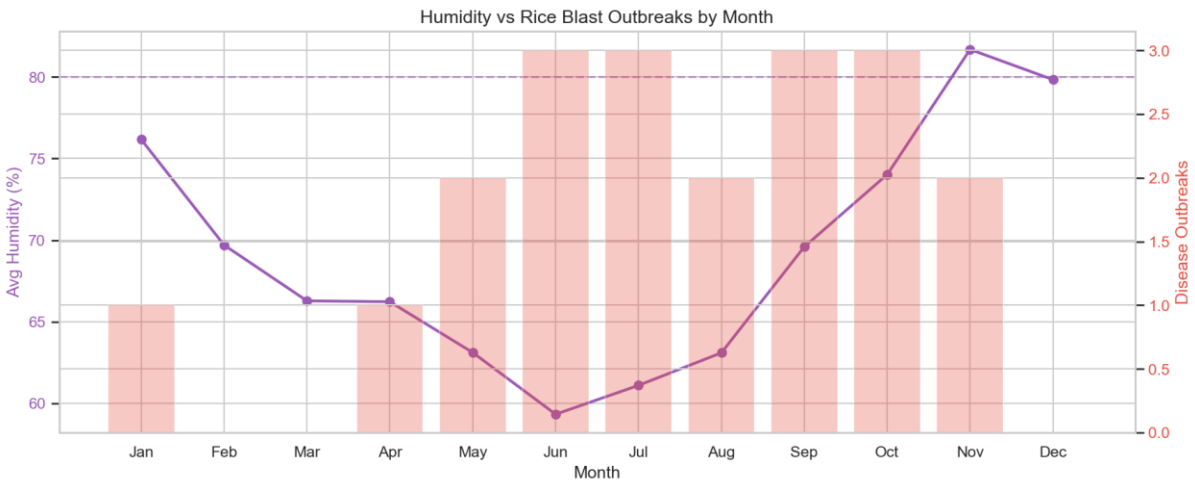
The model was able to classify crop disease risk into Low, Medium, and High categories with good accuracy. The feature importance results show that humidity and rainfall are the most important factors influencing crop disease outbreaks.

8.6 Feature Importance



Feature importance graph shows that humidity, rainfall, and month are the most important features influencing crop disease prediction.

8.7 Relationship Graph



The graph shows that disease outbreaks increase when humidity levels are high, especially above 80%.

8.8 Discussion

From the results, it is clear that climate conditions play a major role in crop disease outbreaks. High humidity and heavy rainfall create favorable conditions for fungal diseases such as Rice Blast and Brown Spot. The machine learning model was able to learn patterns from historical data and predict disease risk effectively.

This system can be used as an early warning system for farmers. By predicting disease risk in advance, farmers can take preventive measures such as pesticide application, proper irrigation management, and crop monitoring. This can help reduce crop loss and improve agricultural productivity.

9. Tools and Technologies

Various tools and technologies were used in this project for data collection, data analysis, visualization, and machine learning model development. These tools helped in implementing the crop disease prediction system efficiently.

The following tools and technologies were used in this project:

Tool	Purpose
Python	Python was used as the main programming language for data analysis and machine learning model development.
Pandas	Pandas library was used for data preprocessing, data cleaning, and data manipulation.
NumPy	NumPy was used for numerical operations and handling arrays.
Matplotlib	Matplotlib was used to create graphs and visualizations such as line charts and bar charts.
Seaborn	Seaborn was used to create heatmaps and statistical visualizations.
Scikit-learn	Scikit-learn was used for machine learning utilities such as train-test split and model evaluation.
XGBoost	XGBoost algorithm was used as the machine learning model to predict crop disease risk.
Jupyter Notebook	Jupyter Notebook was used for writing and executing Python code and performing data analysis.
GitHub	GitHub was used for version control, project documentation, and collaboration.

These tools helped in completing different stages of the project such as data preprocessing, visualization, model development, and documentation.

10. Conclusion

In this project, a crop disease prediction system was developed using climate data and machine learning techniques. Climate factors such as temperature, humidity, and rainfall were analyzed to understand their impact on crop disease outbreaks. Data preprocessing, exploratory data analysis, and data visualization were performed to identify patterns in the dataset. The XGBoost machine learning algorithm was used to predict crop disease risk as Low, Medium, or High. The model achieved approximately 85% accuracy in predicting disease risk. The results showed that humidity and rainfall are the most important factors affecting crop disease outbreaks.

This project demonstrates how data science and machine learning can be applied in agriculture to predict crop disease outbreaks in advance. The system can be used as an early warning tool to help farmers take preventive measures such as pesticide application, irrigation control, and crop monitoring. This can help reduce crop loss and improve agricultural productivity.

11. Future Work

Although the current system is able to predict crop disease risk with good accuracy, there are several improvements that can be made in the future to enhance the system.

- Real-time weather data can be used to make live predictions of crop disease risk.
- A mobile application can be developed so that farmers can easily access disease predictions.
- More machine learning and deep learning algorithms can be tested to improve prediction accuracy.
- The system can be expanded to support multiple crops and different regions.
- The model can be deployed on the cloud to provide real-time disease prediction services.
- Additional features such as soil moisture and pesticide usage data can be included to improve the model.

These improvements can make the system more useful and practical for real-world agricultural applications.

12. References

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